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(12) United States Patent
Sone**(10) Patent No.: US 12,415,872 B2****(45) Date of Patent: Sep. 16, 2025****(54) RUBBER COMPOSITION AND TIRE****(71) Applicant: BRIDGESTONE CORPORATION,**
Tokyo (JP)**(72) Inventor: Ryota Sone, Tokyo (JP)****(73) Assignee: BRIDGESTONE CORPORATION,**
Tokyo (JP)**(*) Notice:** Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 701 days.**(21) Appl. No.: 17/776,479****(22) PCT Filed: Sep. 23, 2020****(86) PCT No.: PCT/JP2020/035868**§ 371 (c)(1),
(2) Date: May 12, 2022**(87) PCT Pub. No.: WO2021/106337**PCT Pub. Date: **Jun. 3, 2021****(65) Prior Publication Data**

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(2013.01); **C08K 3/04** (2013.01); **C08K 3/36**
(2013.01)**(58) Field of Classification Search**CPC .. C08C 19/25; C08K 3/36; C08K 3/04; B60C
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See application file for complete search history.

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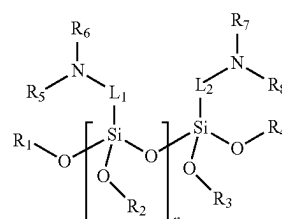
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Primary Examiner — Doris L Lee**(74) Attorney, Agent, or Firm** — Sughrue Mion, PLLC**(57)****ABSTRACT**

Provided is a rubber composition that realizes excellent low heat generating properties while maintaining good steering stability when applied to a tire. The rubber composition of the present disclosure contains a rubber component containing a copolymer having a conjugated diene unit and an aromatic vinyl unit, and a filler, where the copolymer is a modified copolymer having a content of the aromatic vinyl unit of less than 10 mass % and modified with a modifier containing a compound represented by the formula (1).



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12 Claims, No Drawings

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RUBBER COMPOSITION AND TIRE

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a National Stage of International Application No. PCT/JP2020/035868 filed Sep. 23, 2020, claiming priority based on Japanese Patent Application No. 2019-217351 filed Nov. 29, 2019.

TECHNICAL FIELD

This disclosure relates to a rubber composition and a tire.

BACKGROUND

In recent years, the demand for fuel efficiency of automobiles is increasing because of the movement of global carbon dioxide emission regulations caused by the social demand for energy saving and the growing interest in environmental problems. To meet such demand for fuel efficiency, it has been required to reduce the rolling resistance in terms of tire performance. A method of optimizing a tire structure has been studied as a method for reducing the rolling resistance of a tire, but a general method for reducing rolling resistance is to use a material with low heat generating properties as a rubber composition.

Examples of techniques of improving the low heat generating properties of a rubber composition include the one described in JP4478262B (PTL 1), where a rubber composition uses a modified copolymer as a rubber component, and the modified copolymer is obtained by reacting a specific imino group-containing hydrocarbyloxysilane compound.

Because the dispersibility of filler is improved in the rubber composition of PTL 1, a certain rolling resistance-reducing effect can be obtained when the rubber composition is applied to a tire. However, the improvement effect of the rubber composition of PTL 1 is insufficient to meet the recent demand for fuel efficiency, and it is desired to further improve the low heat generating properties.

Techniques of adjusting rubber components have also been developed to improve the low heat generating properties of a rubber composition. For example, JP2018131560A (PTL 2) describes a rubber composition using a styrene butadiene rubber having a reduced amount of styrene as a rubber component.

However, in the case of using a styrene butadiene rubber having a reduced amount of styrene, it is considered that, although the low heat generating properties can be improved, the performance such as steering stability when the rubber composition is applied to a tire is deteriorated.

Further, it is conceivable to reduce the content of filler such as carbon black to improve the low heat generating properties of a rubber composition. However, as in the case of using a styrene butadiene rubber having a reduced amount of styrene, it is considered that the performance such as the reinforcing properties of the rubber, the wear resistance, and the steering stability when applied to a tire is deteriorated.

Therefore, it is required to develop a technique that can greatly improve the low heat generating properties without deteriorating the performance other than the low heat generating properties such as the steering stability when applied to a tire.

CITATION LIST

Patent Literature

PTL 1: JP4478262B

PTL 2: JP2018131560A

SUMMARY

Technical Problem

It could thus be helpful to provide a rubber composition that realizes excellent low heat generating properties while maintaining good steering stability when applied to a tire. It is also helpful to provide a tire with improved rolling resistance and steering stability.

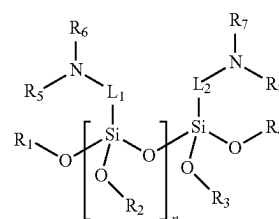
Solution to Problem

As a result of studies to solve the above problems, we found that, by using a copolymer having a conjugated diene unit and an aromatic vinyl unit as a rubber component and setting the content of the aromatic vinyl unit of the copolymer to a small amount of less than 10 mass %, the low heat generating properties of a rubber composition can be improved. However, when the content of the aromatic vinyl unit of the copolymer is low, there is a problem that the performance such as steering stability when the rubber composition is applied to a tire is deteriorated, as described above. Therefore, as a result of further diligent research, we found that the dispersibility of filler can be greatly improved by modifying the copolymer with a specific modifier having oligosiloxane and a tertiary amino group, thereby obtaining, in addition to an effect of improving the low heat generating properties by polymer improvement, further improvement in low heat generating properties and improvement in steering stability when applied to a tire because of improved dispersibility of filler.

We thus provide the following.

The rubber composition of the present disclosure is a rubber composition containing a rubber component containing a copolymer having a conjugated diene unit and an aromatic vinyl unit, and a filler, wherein

the copolymer is a modified copolymer having a content of the aromatic vinyl unit of less than 10 mass % and modified with a modifier containing a compound represented by the formula (1),



where R1 to R8 are each an independent alkyl group having 1 to 20 carbon atoms; L1 and L2 are each an independent alkylene group having 1 to 20 carbon atoms; and n is an integer of 2 to 4.

With the above configuration, it is possible to realize excellent low heat generating properties while maintaining good steering stability when applied to a tire.

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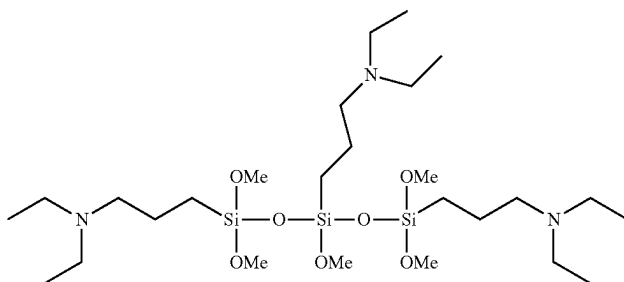
The rubber composition of the present disclosure preferably further contains a conjugated diene-based rubber different from the modified copolymer. This can further improve the steering stability and wear resistance when applied to a tire.

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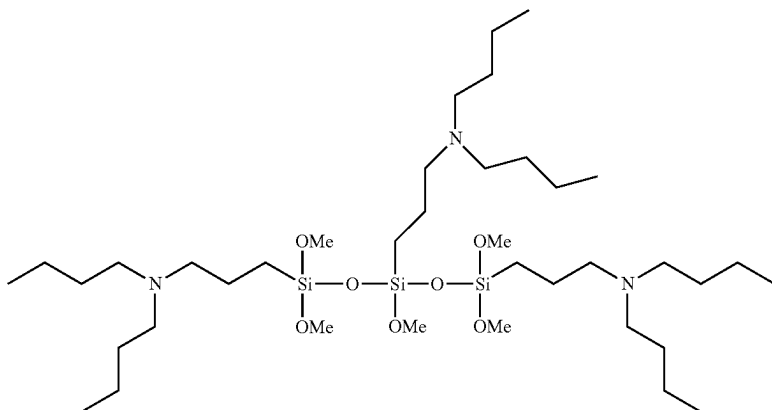
For the rubber composition of the present disclosure, the content of the aromatic vinyl unit in the copolymer is preferably 8 mass % or less. This can provide better low heat generating properties.

For the rubber composition of the present disclosure, the modifier is preferably any one of the formulas (1a) to (1e).

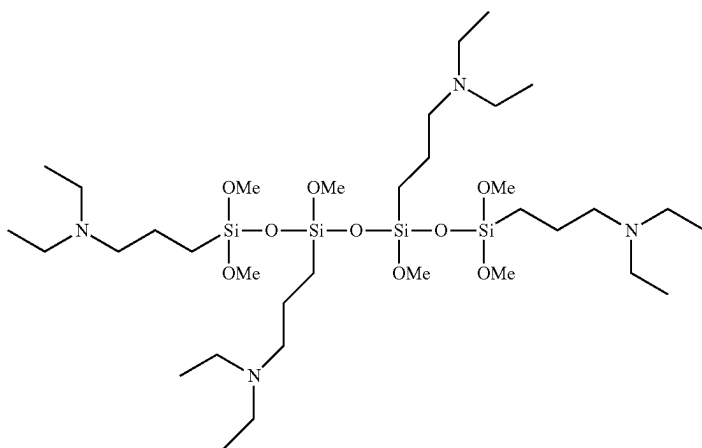
(1a)



(1b)



(1c)

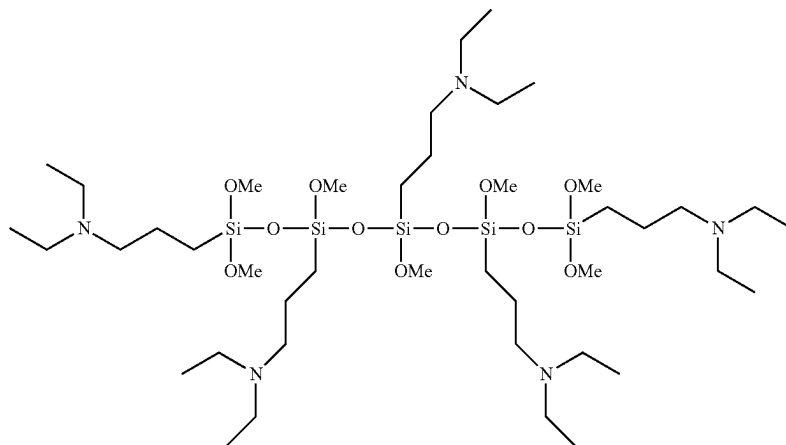


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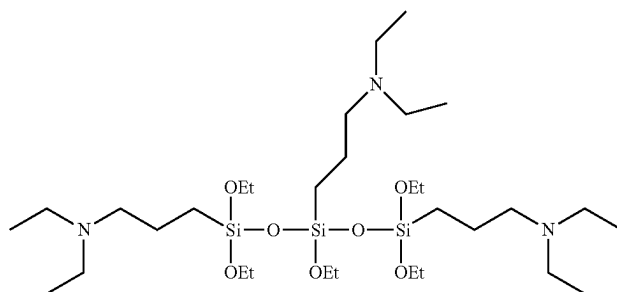
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(1d)

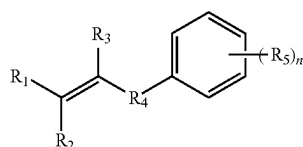


(1e)



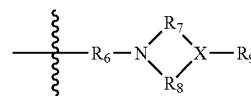
This can realize both the low heat generating properties and the steering stability when applied to a tire at a higher level.

The copolymer is preferably a modified copolymer further modified with a modifier containing a compound represented by the formula (2),



where in the formula (2), R_1 to R_3 are each independently hydrogen; an alkyl group having 1 to 30 carbon atoms; an alkenyl group having 2 to 30 carbon atoms; an alkynyl group having 2 to 30 carbon atoms; a heteroalkyl group having 1 to 30 carbon atoms; a heteroalkenyl group having 2 to 30 carbon atoms; a heteroalkynyl group having 2 to 30 carbon atoms; a cycloalkyl group having 5 to 30 carbon atoms; an aryl group having 6 to 30 carbon atoms; or a heterocyclic group having 3 to 30 carbon atoms, R_4 is a single bond; an alkylene group having 1 to 20 carbon atoms substituted or unsubstituted with a substituent; a cycloalkylene group having 5 to 20 carbon atoms substituted or unsubstituted with a substituent; or an arylene group having 5 to 20 carbon atoms substituted or unsubstituted with a substituent, where the substituent is an alkyl group having 1 to 10 carbon atoms, a cycloalkyl group having 5 to 10 carbon atoms, or an aryl group having 6 to 20 carbon atoms, R_5 is an alkyl group having 1 to 30 carbon atoms; an alkenyl group having

2 to 30 carbon atoms; an alkynyl group having 2 to 30 carbon atoms; a heteroalkyl group having 1 to 30 carbon atoms; a heteroalkenyl group having 2 to 30 carbon atoms; a heteroalkynyl group having 2 to 30 carbon atoms; a cycloalkyl group having 5 to 30 carbon atoms; an aryl group having 6 to 30 carbon atoms; a heterocyclic group having 3 to 30 carbon atoms; or a functional group represented by the following chemical formula (2a) or chemical formula (2b), n is an integer of 1 to 5, when at least one of R_5 is a functional group represented by the following chemical formula (2a) or chemical formula (2b), and n is an integer of 2 to 5, a plurality of R_5 s may be the same as or different from each other,

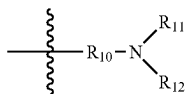


(2a)

where in the formula (2a), R_6 is an alkylene group having 1 to 20 carbon atoms substituted or unsubstituted with a substituent; a cycloalkylene group having 5 to 20 carbon atoms substituted or unsubstituted with a substituent; or an arylene group having 6 to 20 carbon atoms substituted or unsubstituted with a substituent, where the substituent is an alkyl group having 1 to 10 carbon atoms, a cycloalkyl group having 5 to 10 carbon atoms, or an aryl group having 6 to 20 carbon atoms, R_7 and R_8 are each independently an alkyl group having 1 to 10 carbon atoms, a cycloalkyl group having 5 to 10 carbon atoms, or an alkylene group having 1 to 20 carbon atoms substituted or unsubstituted with an aryl group having 6 to 20 carbon atoms, R_9 is hydrogen; an alkyl

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group having 1 to 30 carbon atoms; an alkenyl group having 2 to 30 carbon atoms; an alkynyl group having 2 to 30 carbon atoms; a heteroalkyl group having 1 to 30 carbon atoms; a heteroalkenyl group having 2 to 30 carbon atoms; a heteroalkynyl group having 2 to 30 carbon atoms; a cycloalkyl group having 5 to 30 carbon atoms; an aryl group having 6 to 30 carbon atoms; or a heterocyclic group having 3 to 30 carbon atoms, X is a N, O or S atom, when X is O or S, R₉ does not exist,



where in the formula (2b), R₁₀ is an alkylene group having 1 to 20 carbon atoms substituted or unsubstituted with a substituent; a cycloalkylene group having 5 to 20 carbon atoms substituted or unsubstituted with a substituent; or an arylene group having 6 to 20 carbon atoms substituted or unsubstituted with a substituent, where the substituent is an alkyl group having 1 to 10 carbon atoms, a cycloalkyl group having 5 to 10 carbon atoms, or an aryl group having 6 to 20 carbon atoms, R₁₁ and R₁₂ are each independently an alkyl group having 1 to 30 carbon atoms; an alkenyl group having 2 to 30 carbon atoms; an alkynyl group having 2 to 30 carbon atoms; a heteroalkyl group having 1 to 30 carbon atoms; a heteroalkenyl group having 2 to 30 carbon atoms; a heteroalkynyl group having 2 to 30 carbon atoms; a cycloalkyl group having 5 to 30 carbon atoms; an aryl group having 6 to 30 carbon atoms; or a heterocyclic group having 3 to 30 carbon atoms.

This can realize both the low heat generating properties and the steering stability when applied to a tire at a higher level.

The tire of the present disclosure uses the rubber composition of the present disclosure described above.

With the above configuration, the rolling resistance and the steering stability can be improved.

Advantageous Effect

According to the present disclosure, it is possible to provide a rubber composition that realizes excellent low heat generating properties while maintaining good steering stability when applied to a tire. According to the present disclosure, it is also possible to provide a tire with improved rolling resistance and steering stability.

DETAILED DESCRIPTION

The following describes an embodiment of the present disclosure in detail.

<Rubber Composition>

The rubber composition of the present disclosure is a rubber composition containing a rubber component and a filler.

The following describes each component of the rubber composition of the present disclosure.

(Rubber Component)

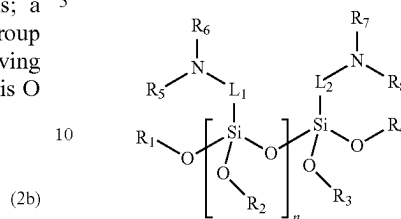
The rubber composition of the present disclosure contains a rubber component.

The rubber component contains a copolymer having a conjugated diene unit and an aromatic vinyl unit, where the

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copolymer is a modified copolymer modified with a modifier containing a compound represented by the formula (1).

(1)



By using a copolymer that has been modified with a modifier containing a compound represented by the formula (1) containing oligosiloxane, which is a filler-affinity functional group, and a tertiary amino group as the rubber component, the dispersibility of the filler in the rubber composition such as silica can be enhanced. As a result, the low heat generating properties are greatly improved and the dispersibility of the filler is improved in the rubber composition of the present disclosure, and other physical properties such as the reinforcing properties, the steering stability when applied to a tire, and the processability can also be improved.

In the formula (1), R₁ to R₈ are each an independent alkyl group having 1 to 20 carbon atoms; L₁ and L₂ are each an independent alkylene group having 1 to 20 carbon atoms; and n is an integer of 2 to 4.

Specifically, in the formula (1), R₁ to R₄ may each independently be a substituted or unsubstituted alkyl group having 1 to 20 carbon atoms. When the R₁ to R₄ are substituted, they may each independently be substituted with at least one substituent selected from the group consisting of an alkyl group having 1 to 10 carbon atoms, a cycloalkyl group having 3 to 10 carbon atoms, an alkoxy group having 1 to 10 carbon atoms, a cycloalkoxy group having 4 to 10 carbon atoms, an aryl group having 6 to 12 carbon atoms, an aryloxy group having 6 to 12 carbon atoms, an alkanoyloxy group having 2 to 12 carbon atoms (alkanoyl, RaCOO⁻, where Ra is an alkyl group having 1 to 9 carbon atoms), an aralkyloxy group having 7 to 13 carbon atoms, an arylalkyl group having 7 to 13 carbon atoms, and an alkylaryl group having 7 to 13 carbon atoms.

More specifically, the R₁ to R₄ may be a substituted or unsubstituted alkyl group having 1 to 10 carbon atoms. More specifically, the R₁ to R₄ may each independently be a substituted or unsubstituted alkyl group having 1 to 6 carbon atoms.

In the formula (1), R₅ to R₈ are each independently a substituted or unsubstituted alkyl group having 1 to 20 carbon atoms. Specifically, they may each independently be a substituted or unsubstituted alkyl group having 1 to 10 carbon atoms. More specifically, they may each independently be a substituted or unsubstituted alkyl group having 1 to 6 carbon atoms. When substituted, they can be substituted with a substituent as described above for R₁ to R₄. When R₅ to R₈ are not an alkyl group but a hydrolyzable substituent, the bonds of N—R₅R₆ and N—R₇R₈ may be hydrolyzed to N—H in the presence of water, which may adversely affect the processability of the polymer.

More specifically, in the compound represented by the formula (1), R₁ to R₄ may be a methyl group or an ethyl group, and R₅ to R₈ may be an alkyl group having 1 to 10 carbon atoms.

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In the present disclosure, the amino group in the compound represented by the formula (1), that is, N—R₅R₆ and N—R₇R₈ are preferably a tertiary amino group. The tertiary amino group provides better processability when the compound represented by the formula (1) is used as a modifier.

Note that when a protecting group for protecting an amino group is bonded to the R₅ to R₈, or hydrogen is bonded to the R₅ to R₈, it may be difficult to obtain the effect of the compound represented by the formula (1). When hydrogen is bonded, anions react with the hydrogen during the modification and lose their reactivity, rendering the modification reaction itself impossible. When a protecting group is bonded, the modification reaction will proceed, but it is deprotected by hydrolysis during subsequent processing in a state of being bonded to the polymer terminal, resulting in a primary or secondary amino group. The deprotected primary or secondary amino group may cause a high viscosity phenomenon in the product during subsequent composition and may cause a decrease in processability.

In the compound represented by the formula (1), L₁ and L₂ are each independently a substituted or unsubstituted alkylene group having 1 to 20 carbon atoms.

More specifically, L₁ and L₂ are each independently an alkylene group having 1 to 10 carbon atoms. More specifically,

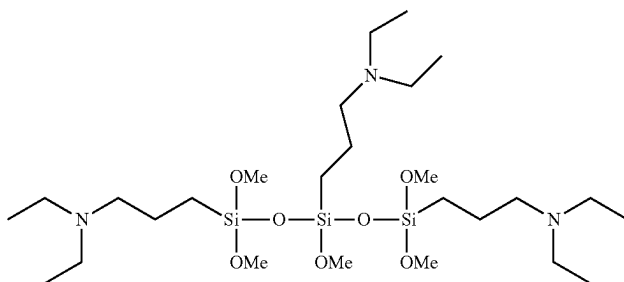
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they may be an alkylene group having 1 to 6 carbon atoms such as a methylene group, an ethylene group, or a propylene group.

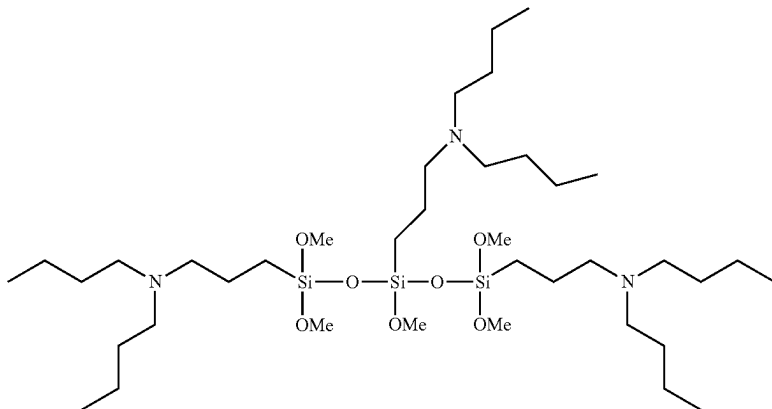
For L₁ and L₂ in the compound represented by the formula (1), the effect improves as the distance between the Si atom and the N atom in the molecule decreases. However, if Si is directly bonded to N, the bond between Si and N may be broken during subsequent processing, and a secondary amino group formed in this case is likely to be washed away by water during subsequent processing. Then, it is difficult to bond a modified copolymer thus obtained to a silica filler through a member of the amino group that promotes the bond with a silica filler, which may deteriorate the dispersion effect of a dispersant. Considering the improvement effect by the length of the bond between Si and N, the L₁ and L₂ are more preferably each independently an alkylene group having 1 to 3 carbon atoms such as a methylene group, an ethylene group, or a propylene group, and more specifically, they may be a propylene group. Further, L₁ and L₂ may be substituted with a substituent as described above for R₁ to R₄.

The compound represented by the formula (1) is preferably, for example, any one of the compounds represented by the following formulas (1a) to (1e). This can achieve both the low heat generating properties and the steering stability when applied to a tire at a higher level.

(1a)



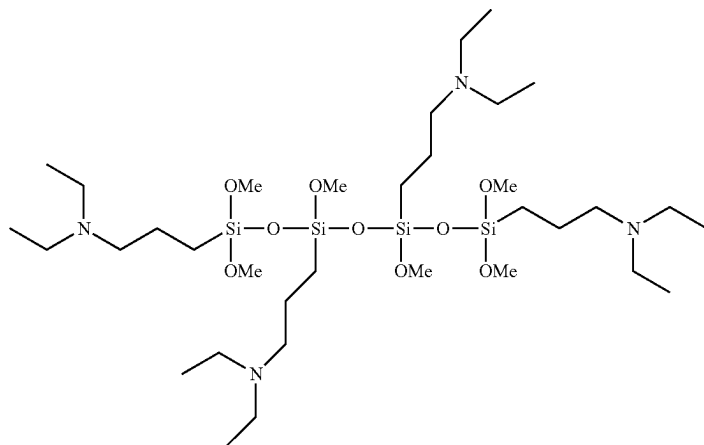
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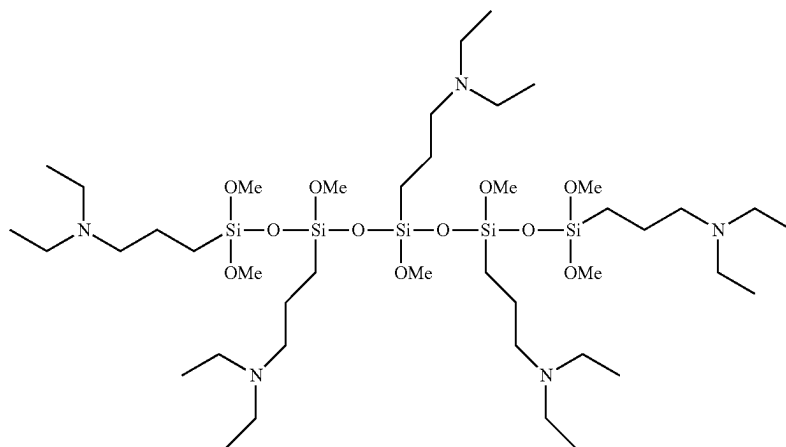
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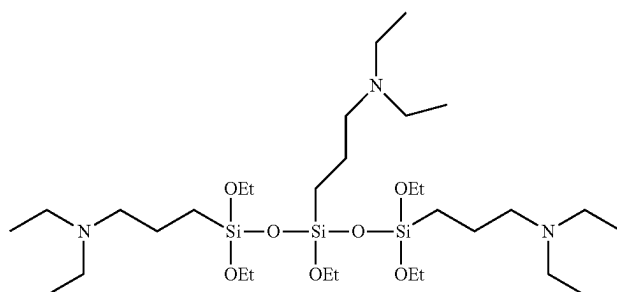
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(1c)



(1d)



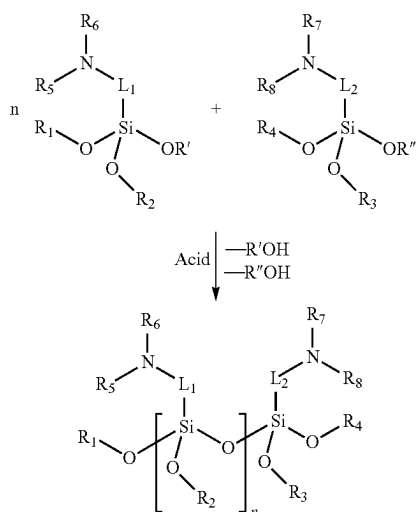
(1e)

In the compound represented by the formula (1) of the
 modifier of the present disclosure, an alkoxy silane structure
 is bonded to an activated terminal of the conjugated diene-
 based polymer, and a Si—O—Si structure and three or more
 amino groups bonded to a terminal exhibit affinity for a filler
 such as silica. In this way, the bond between the filler and the
 modified copolymer can be promoted as compared with a
 conventional modifier containing one amino group in the
 molecule. Further, the degree of bonding of the activated
 terminal of the conjugated diene-based polymer is uniform,
 and it is found that, when observing the change in molecular
 weight distribution before and after coupling, the molecular
 weight distribution after coupling does not increase as

55 compared to before coupling and is kept constant. Therefore,
 the physical properties of the modified copolymer itself are
 not deteriorated, the aggregation of the filler in the rubber
 composition can be prevented, and the dispersibility of the
 filler can be improved. As a result, the processability of the
 rubber composition can be improved. These effects can
 particularly improve the fuel efficiency properties, wear
 properties and steering stability in a well-balanced manner
 when the rubber composition is applied to a tire.

65 Note that the compound represented by the formula (1)
 can be obtained through a condensation reaction represented
 by the following reaction formula 1.

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In the reaction formula 1, R₁ to R₈, L₁ and L₂, and n are the same as those defined in the formula (1) above, and R' and R'' are arbitrary substituents that do not affect the condensation reaction. For example, the R' and R'' may each independently be the same as any one of R₁ to R₄.

The reaction of the reaction formula 1 is carried out under acid conditions, and the acid is not limited if it is a common one for a condensation reaction. A person skill in the art can select an optimum acid according to all kinds of process variables such as the type of reactor in which the reaction is carried out, the starting material, and the reaction temperature.

The copolymer modified with a modifier containing a compound represented by the formula (1) is a copolymer having a conjugated diene unit and an aromatic vinyl unit.

The conjugated diene unit and the aromatic vinyl unit can be randomly arranged and bonded to obtain a random copolymer.

In the rubber composition of the present disclosure, it is required that the content of the aromatic vinyl unit of the copolymer (the mass occupied by the aromatic vinyl unit with respect to the total mass of the copolymer) be less than 10 mass %. This can improve the low heat generating properties of the rubber composition. From the same viewpoint, the content of the aromatic vinyl unit is preferably 8 mass % or less and more preferably 7 mass % or less. From the viewpoint of maintaining a good level of steering stability and wear resistance when the rubber composition is applied to a tire, the content of the aromatic vinyl unit of the copolymer is preferably 3 mass % or more.

The type of a conjugated diene-based monomer as the conjugated diene unit is not particularly limited. For example, it may be at least one selected from the group consisting of 1,3-butadiene, 2,3-dimethyl-1,3-butadiene, piperylene, 3-butyl-1,3-octadiene, isoprene, and 2-phenyl-1,3-butadiene.

The type of an aromatic vinyl monomer as the aromatic vinyl unit is not particularly limited. For example, it may be at least one selected from the group consisting of styrene, α -methylstyrene, 3-methylstyrene, 4-methylstyrene, 4-propylstyrene, 1-vinylnaphthalene, 4-cyclohexylstyrene, 4-(p-methylphenyl) styrene, and 1-vinyl-5-hexylnaphthalene.

The copolymer of the modified copolymer may be a combination of the above-mentioned conjugated diene-based monomer and aromatic vinyl monomer, among which

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styrene butadiene rubber is preferable. This can more reliably realize excellent low heat generating properties without deteriorating other performance and obtain excellent wet performance when applied to a tire.

The modified copolymer may have a narrow molecular weight distribution Mw/Mn (also referred to as polydispersity index (PDI)) of 1.1 to 3.0. When the molecular weight distribution of the modified copolymer exceeds 3.0 or is less than 1.1, the tensile properties and the viscoelasticity may deteriorate when applied to the rubber composition. Considering the remarkable effect of improving the tensile properties and the viscoelasticity of the polymer by controlling the molecular weight distribution of the modified copolymer, the molecular weight distribution of the modified copolymer is preferably 1.3 to 2.0. Note that the modified copolymer has a similar molecular weight distribution to that of the copolymer before being modified with the modifier.

The molecular weight distribution of the modified copolymer can be calculated from a ratio (Mw/Mn) of the weight average molecular weight (Mw) to the number average molecular weight (Mn). The number average molecular weight (Mn) is a common average of individual polymer molecular weights obtained by measuring the molecular weights of n polymer molecules and dividing the total of these molecular weights by n, and the weight average molecular weight (Mw) represents a molecular weight distribution of the polymer composition. The average of total molecular weight can be expressed in grams per mole (g/mol).

In the present disclosure, the weight average molecular weight and the number average molecular weight each are a polystyrene-equivalent molecular weight analyzed by gel permeation chromatography (GPC).

The modified copolymer satisfies the above-mentioned molecular weight distribution conditions and has a number average molecular weight (Mn) of 50,000 g/mol to 2,000,000 g/mol, where the number average molecular weight may be, more specifically, 200,000 g/mol to 800,000 g/mol. The modified copolymer has a weight average molecular weight (Mw) of 100,000 g/mol to 4,000,000 g/mol, where the weight average molecular weight may be, more specifically, 300,000 g/mol to 1,500,000 g/mol.

When the weight average molecular weight (Mw) of the modified copolymer is less than 100,000 g/mol or the number average molecular weight (Mn) of the modified copolymer is less than 50,000 g/mol, the tensile properties when applied to the rubber composition may deteriorate. When the weight average molecular weight (Mw) exceeds 4,000,000 g/mol or the number average molecular weight (Mn) exceeds 2,000,000 g/mol, the workability of the rubber composition deteriorates due to the deterioration of the processability of the modified copolymer, which may render kneading difficult and may render it difficult to sufficiently improve the physical properties of the rubber composition.

More specifically, when the modified copolymer of one embodiment of the present disclosure simultaneously satisfies the conditions of weight average molecular weight (Mw) and number average molecular weight (Mn) as well as the molecular weight distribution, it is possible to improve the viscoelasticity and the processability of the rubber composition when applied to a composition made of rubber in a well-balanced manner.

Further, the present disclosure can provide a method of producing the modified copolymer using a modifier containing a compound represented by the formula (1).

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Specifically, the method of producing the modified copolymer may include a step 1) where an aromatic vinyl-based monomer and a conjugated diene-based monomer are polymerized in a hydrocarbon solvent in the presence of an organic alkali metal compound to obtain an active polymer in which an alkali metal is bonded to at least one terminal, and a step 2) where the active polymer is reacted with a modifier containing a compound represented by the chemical formula 1.

The step 1) is a step for obtaining an active polymer in which an alkali metal is bonded to at least one terminal, which can be carried out by polymerizing an aromatic vinyl-based monomer and a conjugated diene-based monomer in a hydrocarbon solvent in the presence of an organic alkali metal compound.

The hydrocarbon solvent is not particularly limited, and it may be, for example, at least one selected from the group consisting of n-pentane, n-hexane, n-heptane, isooctane, cyclohexane, toluene, benzene, and xylene.

The organic alkali metal compound may be used in an amount of 0.1 mmol to 1.0 mmol with respect to 100 g of the whole monomer.

The organic alkali metal compound is not particularly limited, and it may be, for example, at least one selected from the group consisting of methylolithium, ethyllithium, propyllithium, n-butyllithium, s-butyllithium, t-butyllithium, hexyllithium, n-decylolithium, t-octyllithium, phenyllithium, 1-naphthyllithium, n-eicosyllithium, 4-butylphenyllithium, 4-tolylolithium, cyclohexyllithium, 3,5-di-n-heptylcyclohexyllithium, 4-cyclopentyllithium, naphthyl sodium, naphthyl potassium, lithium alkoxide, sodium alkoxide, potassium alkoxide, lithium sulfonate, sodium sulfonate, potassium sulfonate, lithium amide, sodium amide, potassium amide, and lithium isopropylamide.

The polymerization in the step 1) may be carried out by further adding a polar additive, if necessary, where the polar additive can be added in an amount of 0.001 parts by weight to 1.0 part by weight with respect to 100 parts by weight of the whole monomer. Specifically, it can be added in an amount of 0.005 parts by weight to 0.5 parts by weight, and more specifically 0.01 parts by weight to 0.3 parts by weight, with respect to 100 parts by weight of the whole monomer.

The polar additive may be, for example, at least one selected from the group consisting of tetrahydrofuran, dithetrahydrofurylpropane, diethyl ether, cycloamal ether, dipropyl ether, ethylene dimethyl ether, ethylene dimethyl ether, diethyl glycol, dimethyl ether, tertiary butoxyethoxyethane, bis (3-dimethylaminoethyl) ether, (dimethylaminoethyl) ethyl ether, trimethylamine, triethylamine, tripropylamine, and tetramethylethylenediamine.

When a conjugated diene-based monomer and an aromatic vinyl-based monomer are copolymerized using the polar additive in the production method, the difference in reaction rates may be compensated so that the formation of random copolymer is induced to be easy.

The polymerization in the step 1) can be carried out via adiabatic polymerization or isothermal polymerization.

As used herein, the adiabatic polymerization is a polymerization method including a step of charging an organic alkali metal compound and then performing polymerization by self-reaction heat without arbitrarily applying heat, and the isothermal polymerization is a polymerization method in which the organic alkali metal compound is charged and then heat is arbitrarily applied or removed to maintain a constant temperature of the polymer.

The polymerization may be carried out in a temperature range of 20° C. to 200° C., specifically in a temperature

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range of 0° C. to 150° C., and more specifically in a temperature range of 10° C. to 120° C.

The step 2) is a modification reaction step in which the active polymer is reacted with a modifier containing a compound represented by the formula (1) to produce a modified copolymer.

At this time, the modifier containing a compound represented by the formula (1) may be the same as that described above. The compound represented by the formula (1) can be used at a ratio of 0.1 mol to 2.0 mol with respect to 1 mol of the organic alkali metal compound.

Further, the reaction in the step 2) is a modification reaction for introducing a functional group into the polymer, where each reaction may be carried out in a temperature range of 0° C. to 90° C. for 1 minute to 5 hours.

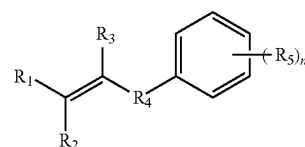
The above-mentioned production method may further include one or more steps of recovering solvent and unreacted monomers and drying after the step 2), if necessary.

To improve the steering stability when applied to a tire, the reinforcing properties, the wear resistance and the like, the rubber component preferably contains a different conjugated diene-based rubber other than the modified copolymer (hereinafter, may be referred to as "other conjugated diene-based rubber").

The other conjugated diene-based rubber can be appropriately selected according to the required performance. For example, it may be a natural rubber (NR) containing cis-1,4-polyisoprene; a modified natural rubber such as epoxidized natural rubber (ENR), deproteinized natural rubber (DPNR), and hydrogenated natural rubber obtained by modifying or purifying the above-mentioned common natural rubber; a synthetic rubber such as styrene-butadiene copolymer (SBR), polybutadiene (BR), and polyisoprene (IR) ethylene-propylene copolymer rubber; or a mixture of any one or more of the above.

The content of the modified copolymer modified with a modifier containing a compound represented by the formula (1) in the rubber component is not particularly limited, and it may be 0.1 mass % to 100 mass %, preferably 10 mass % to 100 mass %, and more preferably 20 mass % to 90 mass %. When the content of the modified copolymer is 0.1 weight % or more, the low heat generating properties can be improved while keeping other physical properties good. As a result, a formed product manufactured using the rubber composition such as a tire can more reliably obtain the effects such as fuel efficiency properties, wear properties and braking properties.

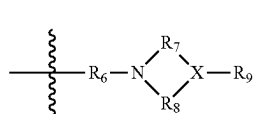
As described above, the copolymer is modified with a modifier containing a compound represented by the formula (1), but it is preferably further modified with a modifier containing a compound represented by the formula (2). This can further improve the dispersibility of the filler in the rubber composition, so that both the low heat generating properties and the steering stability when applied to a tire can be achieved at a higher level, and the wear resistance and the processability can be further improved.



(2)

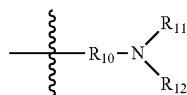
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In the formula (2), R_1 to R_3 are each independently hydrogen; an alkyl group having 1 to 30 carbon atoms; an alkenyl group having 2 to 30 carbon atoms; an alkynyl group having 2 to 30 carbon atoms; a heteroalkyl group having 1 to 30 carbon atoms; a heteroalkenyl group having 2 to 30 carbon atoms; a heteroalkynyl group having 2 to 30 carbon atoms; a cycloalkyl group having 5 to 30 carbon atoms; an aryl group having 6 to 30 carbon atoms; or a heterocyclic group having 3 to 30 carbon atoms, R_4 is a single bond; an alkylene group having 1 to 20 carbon atoms substituted or unsubstituted with a substituent; a cycloalkylene group having 5 to 20 carbon atoms substituted or unsubstituted with a substituent, where the substituent is an alkyl group having 1 to 10 carbon atoms, a cycloalkyl group having 5 to 10 carbon atoms, or an aryl group having 6 to 20 carbon atoms, R_5 is an alkyl group having 1 to 30 carbon atoms; an alkenyl group having 2 to 30 carbon atoms; an alkynyl group having 2 to 30 carbon atoms; a heteroalkyl group having 1 to 30 carbon atoms; a heteroalkenyl group having 2 to 30 carbon atoms; a heteroalkynyl group having 2 to 30 carbon atoms; a cycloalkyl group having 5 to 30 carbon atoms; an aryl group having 6 to 30 carbon atoms; a heterocyclic group having 3 to 30 carbon atoms; or a functional group represented by the following chemical formula (2a) or chemical formula (2b), n is an integer of 1 to 5, when at least one of R_5 is a functional group represented by the following chemical formula (2a) or chemical formula (2b), and n is an integer of 2 to 5, a plurality of R_5 s may be the same as or different from each other.



(2a)

In the formula (2a), R_6 is an alkylene group having 1 to 20 carbon atoms substituted or unsubstituted with a substituent; a cycloalkylene group having 5 to 20 carbon atoms substituted or unsubstituted with a substituent; or an arylene group having 6 to 20 carbon atoms substituted or unsubstituted with a substituent, where the substituent is an alkyl group having 1 to 10 carbon atoms, a cycloalkyl group having 5 to 10 carbon atoms, or an aryl group having 6 to 20 carbon atoms, R_7 and R_8 are each independently an alkyl group having 1 to 10 carbon atoms, a cycloalkyl group having 5 to 10 carbon atoms, or an alkylene group having 1 to 20 carbon atoms substituted or unsubstituted with an aryl group having 6 to 20 carbon atoms, R_9 is hydrogen; an alkyl group having 1 to 30 carbon atoms; an alkenyl group having 2 to 30 carbon atoms; an alkynyl group having 2 to 30 carbon atoms; a heteroalkyl group having 1 to 30 carbon atoms; a heteroalkenyl group having 2 to 30 carbon atoms; a heteroalkynyl group having 2 to 30 carbon atoms; a cycloalkyl group having 5 to 30 carbon atoms; an aryl group having 6 to 30 carbon atoms; or a heterocyclic group having 3 to 30 carbon atoms, X is a N, O or S atom, when X is O or S, R_9 does not exist.



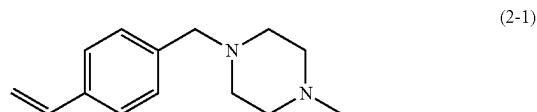
(2b)

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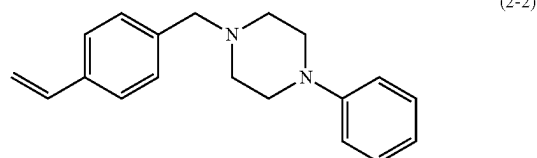
In the formula (2b), R_{10} is an alkylene group having 1 to 20 carbon atoms substituted or unsubstituted with a substituent; a cycloalkylene group having 5 to 20 carbon atoms substituted or unsubstituted with a substituent; or an arylene group having 6 to 20 carbon atoms substituted or unsubstituted with a substituent, where the substituent is an alkyl group having 1 to 10 carbon atoms, a cycloalkyl group having 5 to 10 carbon atoms, or an aryl group having 6 to 20 carbon atoms, R_{11} and R_{12} are each independently an alkyl group having 1 to 30 carbon atoms; an alkenyl group having 2 to 30 carbon atoms; an alkynyl group having 2 to 30 carbon atoms; a heteroalkyl group having 1 to 30 carbon atoms; a heteroalkenyl group having 2 to 30 carbon atoms; a heteroalkynyl group having 2 to 30 carbon atoms; a cycloalkyl group having 5 to 30 carbon atoms; an aryl group having 6 to 30 carbon atoms; or a heterocyclic group having 3 to 30 carbon atoms.

In the compound represented by the formula (2), R_1 to R_3 may be each independently hydrogen; an alkyl group having 1 to 10 carbon atoms; an alkenyl group having 2 to 10 carbon atoms; or an alkynyl group having 2 to 10 carbon atoms, R_4 may be a single bond; or an unsubstituted alkylene group having 1 to 10 carbon atoms; R_5 may be an alkyl group having 1 to 10 carbon atoms; an alkenyl group having 2 to 10 carbon atoms; an alkynyl group having 2 to 10 carbon atoms; or a functional group represented by the following chemical formula (2a) or chemical formula (2b), in the formula (2a), R_6 may be an unsubstituted alkylene group having 1 to 10 carbon atoms, R_7 and R_8 may be each independently an unsubstituted alkylene group having 1 to 10 carbon atoms, R_9 may be an alkyl group having 1 to 10 carbon atoms; a cycloalkyl group having 5 to 20 carbon atoms; an aryl group having 6 to 20 carbon atoms; or a heterocyclic group having 3 to 20 carbon atoms, in the formula (2b), R_{10} may be an unsubstituted alkylene group having 1 to 10 carbon atoms, R_{11} and R_{12} may be each independently an alkyl group having 1 to 10 carbon atoms; a cycloalkyl group having 5 to 20 carbon atoms; an aryl group having 6 to 20 carbon atoms; or a heterocyclic group having 3 to 20 carbon atoms.

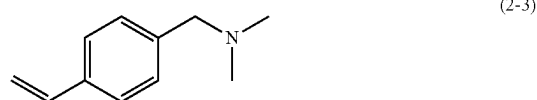
More specifically, the compound represented by the formula (2) may be a compound represented by the following formulas (2-1) to (2-3).



(2-1)



(2-2)



(2-3)

When the copolymer is modified with a modifier containing a compound represented by the formula (2), the modifier containing a compound represented by the formula (2) is used as a modification initiator.

Specifically, by polymerizing a conjugated diene monomer and an aromatic vinyl monomer in the presence of a modifier containing a compound represented by the formula (2) in a hydrocarbon solvent, a modifying group derived from the compound represented by the formula (2) can be added to the copolymer, for example.

The polymerization of the conjugated diene monomer and the aromatic vinyl monomer may be, for example, anionic polymerization.

Specifically, it may be living anionic polymerization having an anionic active site at the polymerization terminal by a growth polymerization reaction by anion, for example.

The polymerization may be temperature-rising polymerization, isothermal polymerization or constant-temperature polymerization (adiabatic polymerization). The constant-temperature polymerization is a polymerization method including a step of charging a modifier containing a compound represented by the formula (2) and then performing polymerization by self-reaction heat without arbitrarily applying heat, the temperature-rising polymerization is a polymerization method in which a modification initiator is charged and then heat is arbitrarily applied to raise the temperature, and the isothermal polymerization is a polymerization method in which the modification initiator is charged and then heat is applied to increase the heat or remove the heat to keep the temperature of the polymer constant.

(Filler)

The rubber composition of the present disclosure contains a filler in addition to the above-described rubber component.

By using a filler together with the rubber component containing the modified copolymer, the dispersibility of the filler is enhanced, the performance such as steering stability, strength, wear resistance, and wet grip properties when the rubber composition is applied to a tire is maintained at a high level, and excellent low heat generating properties can be realized at the same time.

The content of the filler is not particularly limited, but it is preferably 10 parts by mass to 160 parts by mass and more preferably 30 parts by mass to 120 parts by mass with respect to 100 parts by mass of the rubber component. This is because better low heat generating properties and wear resistance can be realized by optimizing the amount of the filler. When the content is 10 parts by mass or more, sufficient wear resistance can be obtained. When the content is 160 parts by mass or less, deterioration of low heat generating properties can be suppressed.

The type of the filler is not particularly limited. For example, carbon black, silica, and other inorganic fillers can be contained. Among the above, the filler preferably contains at least silica. This can further improve the low heat generating properties, the wear resistance, and the wet performance when used in a tire. It can also suppress deterioration of the processability of the composition.

The CTAB (cetyltrimethylammonium bromide) specific surface area of the silica is preferably 50 m²/g or more and preferably 350 m²/g or less. When the CTAB specific surface area of the silica is 50 m²/g or more, the wear resistance is further improved, and when the CTAB specific surface area of the silica is 350 m²/g or less, the rolling resistance is reduced.

The type of the silica is not particularly limited. Examples thereof include wet silica (hydrous silicic acid), dry silica (anhydrous silicic acid), calcium silicate, and aluminum silicate, among which wet silica is preferable. These silicas may be used alone or in combination of two or more.

Further, examples of the silica include wet silica (hydrous silicic acid), dry silica (anhydrous silicic acid), calcium silicate, and aluminum silicate, among which wet silica is preferable. These silicas may be used alone or in combination of two or more.

The wet silica may be precipitated silica. The precipitated silica is silica obtained by aggregating primary particles by, at an initial stage of production, advancing the reaction of a reaction solution in a relatively high temperature and neutral to alkaline pH range to grow silica primary particles and then controlling them to acidic pH range.

The content of the silica is not particularly limited, but it is preferably 10 parts by mass to 160 parts by mass and more preferably 30 parts by mass to 120 parts by mass with respect to 100 parts by mass of the rubber component. This is because better low heat generating properties and wear resistance can be realized by optimizing the amount of the filler. When the content is 10 parts by mass or more, sufficient wear resistance can be obtained. When the content is 30 parts by mass or less, deterioration of low heat generating properties can be suppressed.

The filler preferably contains carbon black in addition to the silica. This can realize better reinforcing properties and wear resistance.

Examples of the carbon black include carbon black of GPF, FEF, SRF, HAF, ISAF, IISAF, and SAF grades.

The content of the carbon black is preferably 2 parts by mass or more and more preferably 4 parts by mass or more with respect to 100 parts by mass of the rubber component, from the viewpoint of obtaining better wear resistance. This is because the wear resistance of the rubber composition can be further improved by setting the content of the carbon black to 2 parts by mass or more with respect to 100 parts by mass of the rubber component. The content of the carbon black is preferably 90 parts by mass or less and more preferably 70 parts by mass or less with respect to 100 parts by mass of the rubber component. This is because, by setting the content of the carbon black to 90 parts by mass or less with respect to 100 parts by mass of the rubber component, it is possible to further improve the low heat generating properties and the processability while maintaining the wear resistance at a high level.

Examples of the other filler include an inorganic compound represented by the following formula (A).



(where M is at least one selected from metals selected from the group consisting of Al, Mg, Ti, Ca and Zr, oxides or hydroxides of these metals, hydrates of these, and carbonates of these metals; and n, x, y and z are an integer of 1 to 5, an integer of 0 to 10, an integer of 2 to 5, and an integer of 0 to 10, respectively.)

Examples of the inorganic compound of the formula (A) include alumina (Al₂O₃) such as γ-alumina and α-alumina; alumina monohydrate (Al₂O₃·H₂O) such as boehmite and diaspore; aluminum hydroxide [Al(OH)₃] such as gibbsite and bayerite; aluminum carbonate [Al₂(CO₃)₃], magnesium hydroxide [Mg(OH)₂], magnesium oxide (MgO), magnesium carbonate (MgCO₃), talc (3MgO·4SiO₂·H₂O), attapulgite (5MgO·8SiO₂·9H₂O), titanium white (TiO₂), titanium black (TiO_{2n-1}), calcium oxide (CaO), calcium hydroxide [Ca(OH)₂], aluminum oxide magnesium (MgO·Al₂O₃), clay (Al₂O₃·SiO₂), kaolin (Al₂O₃·2SiO₂·2H₂O), pyrophyllite (Al₂O₃·4SiO₂·H₂O), bentonite (Al₂O₃·4SiO₂·2H₂O), aluminum silicate (Al₂SiO₅, Al₄·3SiO₄·5H₂O), etc.), magnesium silicate (Mg₂SiO₄, MgSiO₃, etc.), calcium silicate (Ca₂SiO₄, etc.), aluminum

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calcium silicate ($\text{Al}_2\text{O}_3\text{CaO} \cdot 2\text{SiO}_2$, etc.), calcium magnesium silicate (CaMgSiO_4), calcium carbonate (CaCO_3), zirconium oxide (ZrO_2), zirconium hydroxide [$\text{ZrO}(\text{OH})_2 \cdot n\text{H}_2\text{O}$], zirconium carbonate [$\text{Zr}(\text{CO}_3)_2$], and crystalline aluminosilicate containing hydrogen, alkali metal or alkaline earth metal that corrects the charge like various zeolites.

(Other Components)

The rubber composition of the present disclosure may contain other components in addition to the above-described rubber component and filler if the effect of the present disclosure is not impaired.

Examples of the other component include silane coupling agent, thermoplastic resin, plasticizer, liquid rubber, age resistor, crosslinking accelerator, crosslinking agent, crosslinking promoting aid, antiozonant, and surfactant, and additives that are commonly used in the rubber industry can be appropriately contained.

When silica is contained as the filler, the dispersibility of the silica is enhanced and better low heat generating properties and reinforcing properties can be obtained by using the silane coupling agent together with the silica.

The silane coupling agent is not particularly limited. Examples thereof include bis (3-triethoxysilylpropyl) tetrasulfide, bis (3-triethoxysilylpropyl) trisulfide, bis (3-triethoxysilylpropyl) disulfide, bis (2-triethoxysilylethyl) tetrasulfide, bis (3-trimethoxysilylpropyl) tetrasulfide, bis (2-trimethoxysilylethyl) tetrasulfide, 3-mercaptopropyltrimethoxysilane, 3-mercaptopropyltriethoxysilane, 2-mercaptoethyltrimethoxysilane, 2-mercaptoethyltriethoxysilane, 3-trimethoxysilylpropyl-N,N-dimethylthiocarbamoyl tetrasulfide, 3-triethoxysilylpropyl-N,N-dimethylthiocarbamoyl tetrasulfide, 2-triethoxysilylethyl-N,N-dimethylthiocarbamoyl tetrasulfide, 3-trimethoxysilylpropylbenzothiazolyl tetrasulfide, 3-triethoxysilylpropylbenzoyl tetrasulfide, 3-triethoxysilylpropyl methacrylate monosulfide, 3-trimethoxysilylpropyl methacrylate monosulfide, bis (3-diethoxymethylsilylpropyl) tetrasulfide, 3-mercaptopropyl dimethoxymethylsilane, dimethoxymethylsilylpropyl-N,N-dimethylthiocarbamoyl tetrasulfide, and dimethoxymethylsilylpropylbenzothiazolyl tetrasulfide. A mixture of any one or more of these can also be used.

From the viewpoint of improving the reinforcing properties of the rubber composition, it is preferable to contain bis (3-triethoxysilylpropyl) polysulfide or 3-trimethoxysilylpropylbenzothiazyl tetrasulfide as the silane coupling agent.

Because the rubber composition of the present disclosure uses a modified copolymer where a functional group having a high affinity with a silica-based filler has been introduced into an active site as the rubber component, the blending amount of the silane coupling agent can be reduced as compared with common cases. For example, the content of the silane coupling agent may be 1 part by mass to 20 parts by mass with respect to 100 parts by mass of the silica. When used in the above range, it is possible to sufficiently exhibit the effect as a coupling agent and prevent gelation of the rubber component at the same time. From the same viewpoint, the content of the silane coupling agent may be 5 parts by mass to 15 parts by mass with respect to 100 parts by mass of the silica.

The rubber composition of the present disclosure may further contain a thermoplastic resin. By containing the thermoplastic resin, the processability of the rubber composition can be improved, and additionally, it is possible to improve the braking performance on dry road surfaces and wet road surfaces when the rubber composition is used for a tire.

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The type of the thermoplastic resin is not particularly limited. Examples thereof include C5-based resin, C9-based resin, C5 to C9-based resin, dicyclopentadiene-based resin, rosin-based resin, alkyl phenol-based resin, and terpene phenol-based resin.

The C5-based resin refers to a C5-based synthetic petroleum resin and refers to a solid polymer obtained by polymerizing a C5 fraction using a Friedel-Crafts catalyst such as AlCl_3 or BF_3 . Specific examples thereof include a copolymer containing isoprene, cyclopentadiene, 1,3-pentadiene, 1-pentene and the like as main components, a copolymer of 2-pentene and dicyclopentadiene, and a polymer mainly composed of 1,3-pentadiene.

The C9-based resin refers to a C9-based synthetic petroleum resin and refers to a solid polymer obtained by polymerizing a C9 fraction using a Friedel-Crafts catalyst such as AlCl_3 or BF_3 . Specific examples thereof include a copolymer containing indene, methylindene, α -methylstyrene, vinyltoluene and the like as main components.

The C5 to C9-based resin refers to a C5 to C9-based synthetic petroleum resin and refers to a solid polymer obtained by polymerizing a C5 to C9 fraction using a Friedel-Crafts catalyst such as AlCl_3 or BF_3 . Examples thereof include a copolymer containing styrene, vinyltoluene, α -methylstyrene, indene and the like as main components. In the present disclosure, the C5 to C9-based resin is preferably a resin having a small amount of C9 or more components from the viewpoint of the compatibility with the rubber component. As used herein, the "small amount of C9 or more components" means that the C9 or more components in the total amount of the resin are less than 50 mass % and preferably 40 mass % or less.

The dicyclopentadiene-based resin is a petroleum resin using dicyclopentadiene in the above-mentioned C5 fraction as a main raw material. Examples thereof include products of "MARUKAREZ M" Series (M-890A, M-845A, M-990A, etc.) of Maruzen Petrochemical Co., Ltd.

The rosin-based resin may be a natural resin rosin such as gum rosin contained in crude turpentine and tall oil, tall oil rosin, and wood rosin, and it may be modified rosin, rosin derivative, or modified rosin derivative such as polymerized rosin, and partially hydrogenated rosin thereof; glycerin ester rosin, and partially hydrogenated rosin and fully hydrogenated rosin thereof; and pentaerythritol ester rosin, and partially hydrogenated rosin and polymerized rosin thereof.

The alkyl phenol-based resin is a phenol-based resin having an alkyl group. Examples thereof include an alkyl phenol-acetylene resin such as p-tert-butylphenol-acetylene resin, and an alkyl phenol-formaldehyde resin having a low degree of polymerization.

The terpene phenol-based resin is a resin that can be obtained by reacting terpenes with various phenols using a Friedel-Crafts catalyst, or by further performing condensation with formalin. The terpenes as a raw material are not particularly limited, but they are preferably monoterpene hydrocarbons such as α -pinene and limonene, more preferably those containing α -pinene, and particularly preferably α -pinene. A terpene phenol-based resin having a large proportion of phenol components is suitable for the present disclosure. These resins may be used alone or in combination of two or more.

Further, a novolak-type phenol resin is preferably contained as the phenol resin. By containing the novolak-type phenol resin, the elastic modulus in the rubber composition can be increased and the steering stability can be improved without using a curing agent and without deteriorating the wet performance.

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The content of the thermoplastic resin is not particularly limited. However, from the viewpoint of improving the processability and the braking properties when applied to a tire without deteriorating the wear resistance and the reinforcing properties, it is preferably 3 parts by mass to 50 parts by mass and more preferably 5 parts by mass to 30 parts by mass with respect to 100 parts by mass of the rubber component.

The age resistor may be a known one, which is not particularly limited. Examples thereof include a phenol-based age resistor, an imidazole-based age resistor, and an amine-based age resistor. These age resistors may be used alone or in combination of two or more.

The crosslinking accelerator may be a known one, which is not particularly limited. Examples thereof include a thi-azole-based vulcanization accelerator such as 2-mercapto-benzothiazole, dibenzothiazyl disulfide; a sulfenamide-based vulcanization accelerator such as N-cyclohexyl-2-benzothiazyl sulfenamide and N-t-butyl-2-benzothiazyl sulfenamide; a guanidine-based vulcanization accelerator such as diphenyl guanidine; a thiuram-based vulcanization accelerator such as tetramethyl thiuram disulfide, tetraethyl thiuram disulfide, tetrabutyl thiuram disulfide, tetradodecyl thiuram disulfide, tetraoctyl thiuram disulfide, tetrabenzyl thiuram disulfide, and dipentamethylene thiuram tetrasul- fide; a dithiocarbamate-based vulcanization accelerator such as zinc dimethyldithiocarbamate; and zinc dialkyldithio- phosphate. These crosslinking accelerators may be used alone or in combination of two or more.

The crosslinking agent is not particularly limited, either. Examples thereof include sulfur and a bismaleimide com- pound. These crosslinking agents may be used alone or in combination of two or more.

Examples of the types of the bismaleimide compound include N,N'-o-phenylene bismaleimide, N,N'-m-phenylene bismaleimide, N,N'-p-phenylene bismaleimide, N,N'-(4,4'-diphenylmethane) bismaleimide, 2,2-bis-[4-(4-maleimide- phenoxy)phenyl]propane, and bis(3-ethyl-5-methyl-4-male- imidephenyl) methane. In the present disclosure, N,N'-m- phenylene bismaleimide, N,N'-(4,4'-diphenylmethane) bismaleimide and the like may be suitably used.

Examples of the crosslinking promoting aid include zinc oxide (ZnO) and a fatty acid. The fatty acid may be a saturated or unsaturated, linear or branched fatty acid. The carbon number of the fatty acid is not particularly limited, and it may be a fatty acid with 1 to 30, preferably 15 to 30 carbon atoms, for example. Specific examples thereof include naphthenic acids such as cyclohexanoic acid (cyclo- hexanecarboxylic acid) and alkylcyclopentane with side chains; saturated fatty acids such as hexanoic acid, octanoic acid, decanoic acid (including branched carboxylic acids such as neodecanoic acid), dodecanoic acid, tetradecanoic acid, hexadecanoic acid, and octadecanoic acid (stearic acid); unsaturated fatty acids such as methacrylic acid, oleic acid, linoleic acid, and linolenic acid; resin acids such as rosin, tall oil acid, and abietic acid. These may be used alone or in combination of two or more. In the present disclosure, zinc oxide or stearic acid may be suitably used.

A method of producing the rubber composition of the present disclosure is not particularly limited, and the rubber composition can be obtained by blending and kneading all the components (rubber component, filler, and other com- ponents) of the rubber composition.

<Tire>

A tire of the present disclosure uses the rubber composi- tion of the present disclosure described above. By using the rubber composition of the present disclosure as a tire mate-

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rial, the obtained tire can obtain greatly improved steering stability and rolling resistance.

In the tire of the present disclosure, the above-described rubber composition is specifically applied to a member, and it is particularly preferable to apply the rubber composition to a tread among such tire members. A tire using the rubber composition in a tread can realize a high level of reinforcing properties (and thus wear resistance, steering stability, etc.) in addition to an effect of reducing rolling resistance. Examples of a gas to be filled in the tire of the present disclosure include normal air, air with different oxygen partial pressure, and an inert gas such as nitrogen.

EXAMPLES

The following describes the present disclosure in more detail with reference to examples, but the present disclosure is not limited to the following examples.

Example 1 and Comparative Example 1

Samples of each rubber composition are prepared accord- ing to the chemical composition listed in Table 1. The blending amount of each component is indicated in part by mass with respect to 100 parts by mass of the rubber component.

The "modified SBR-1" and "modified SBR-2" in Table 1 are prepared under the following conditions.

(Preparation of Modified SBR-1)

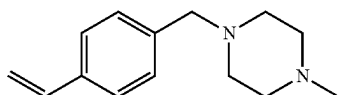
A cyclohexane solution of 1,3-butadiene and a cyclo- hexane solution of styrene are added to a dry, nitrogen- substituted 800 ml pressure-resistant glass container so that 67.5 g of 1,3-butadiene and 7.5 g of styrene are added, and 0.6 mmol of 2,2-ditetrahydrofurylpropane and 0.8 mmol of n-butyllithium are added to the glass container, and then polymerization is carried out at 50° C. for 1.5 hours. At this time, 0.72 mmol of [N,N-bis(trimethylsilyl)-(3-amino-1- propyl)](methyl) (diethoxy) silane is added to the polymer- ization reaction system in which the polymerization conver- sion rate has almost reached 100%, and a modification reaction is carried out at 50° C. for 30 minutes. Next, 2 ml of an isopropanol 5 mass % solution of 2,6-di-t-butyl-p- cresol (BHT) is added to terminate the reaction, and the mixture is dried according to a conventional method to obtain modified SBR-1.

As a result of measuring the microstructure of the obtained modified SBR-1, the bound styrene content is 10 mass %, the vinyl content of the butadiene portion is 40%, and the polystyrene-equivalent peak molecular weight obtained by gel permeation chromatography is 200,000. (Production Example of Modifier)

Two vacuum-dried 4L stainless steel pressure vessels are prepared. To the first pressure vessel, 944 g of cyclohexane, 161 g of a compound represented by the following chemical formula 2-1, and 86 g of tetramethylethylenediamine are charged to prepare a first reaction solution. At the same time, 318 g of liquid 20 weight % n-butyllithium, and 874 g of cyclohexane are charged into the second pressure vessel to prepare a second reaction solution. At this time, the molar ratio of the compound represented by the following chemi- cal formula (2-1), n-butyllithium and tetramethylethylene- diamine is 1:1:1. With the pressure of each pressure vessel maintained at 7 bar, the first reaction solution is injected into a first continuous channel at an injecting rate of 1.0 g/min and the second reaction solution is injected into a second continuous channel at an injecting rate of 1.0 g/min, respec- tively, in a continuous reactor using a mass flow meter. At

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this time, the temperature of the continuous reactor is maintained at -10°C ., the internal pressure is maintained at 3 bar using a back pressure regulator, and the residence time in the reactor is adjusted to be within 10 minutes. The reaction is terminated to obtain a modification initiator.

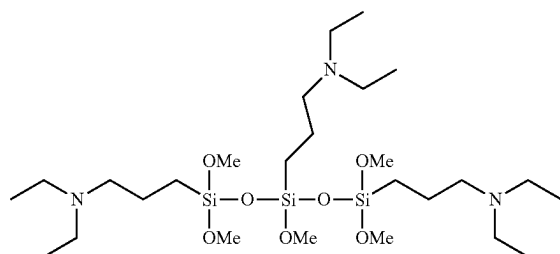


(Preparation of Modified SBR-2)

Continuous reactors in which three reactors are connected in series are prepared, where in the first reactor, a styrene solution in which styrene is dissolved in n-hexane at 60 weight % is injected at a rate of 0.84 kg/h, a 1,3-butadiene solution in which 1,3-butadiene is dissolved in n-hexane at 60 weight % is injected at a rate of 15.10 kg/h, n-hexane is injected at a rate of 47.66 kg/h, a 1,2-butadiene solution in which 1,2-butadiene is dissolved in n-hexane at 2.0 weight % is injected at a rate of 10 g/h, a solution in which 2,2-(di-2(tetrahydrofuryl) propane is dissolved in n-hexane at 10 weight % is injected as a polar additive at a rate of 10.0 g/h, and the modification initiator produced in the above production example is injected at a rate of 292.50 g/h. At this time, the temperature of the first reactor is maintained at 50°C . When the polymerization conversion rate reaches 43%, the polymer is transferred from the first reactor to the second reactor through a transfer pipe.

Subsequently, a 1,3-butadiene solution in which 1,3-butadiene is dissolved in n-hexane at 60 weight % is injected into the second reactor at a rate of 0.68 kg/h. At this time, the temperature of the second reactor is maintained at 65°C . When the polymerization conversion rate is 95% or more, the polymer is transferred from the second reactor to the third reactor through a transfer pipe.

The polymer is transferred from the second reactor to the third reactor, and a solution in which the following formula (1a) is dissolved as a modifier is charged into the third reactor (modifier: act. Li=1:1 mol). The temperature of the third reactor is maintained at 65°C .



Next, an IR1520 (BASF) solution dissolved at 30 weight % as an antioxidant is injected into the polymerization solution discharged from the third reactor at a rate of 170 g/h and stirred. As a result, the obtained polymer is put into warm water heated by steam and stirred, and the solvent is removed to obtain modified SBR-2.

As a result of measuring the microstructure of the obtained modified SBR-2, the styrene content is 5 mass %, and the vinyl content of the butadiene portion is 37%.

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<Evaluation>

The following evaluations are performed on the obtained rubber composition samples of Example 1 and Comparative Example 1. The results are listed in Table 1.

(1) Low Heat Generating Properties

The loss tangent ($\tan \delta$) of each sample is measured under conditions of a temperature of 30°C ., a strain of 5%, and a frequency of 15 Hz using a dynamic viscoelasticity measuring device for high frequency manufactured by Metravig. The obtained value of $\tan \delta$ is indicated as an index with the value of Comparative Example 1 being 100 and is listed in Table 1. The smaller the index value of $\tan \delta$ is, the better the low heat generating properties are.

(2) Steering Stability

The storage shear modulus $G'(\text{Pa})$ of each sample is measured under conditions of 30°C ., a strain of 10%, and a frequency of 15 Hz using a dynamic viscoelasticity measuring device for high frequency manufactured by Metravig. The obtained value of G' is indicated as an index with the value of Comparative Example 1 being 100 and is listed in Table 1. The larger the index value of G' is, the better the steering stability when applied to a tire is.

TABLE 1

		Comparative Example 1	Example 1
Chemical composition	NR *1	50	50
	Modified SBR-1	50	—
	Modified SBR-2	—	50
	Carbon black *2	7	7
	Silica *3	66	66
	Resin *4	10	10
	Processing aid A *16	2	2
	Processing aid B *17	1	1
	Stearic acid *5	1	1
	Zinc oxide *6	3	3
	Oil *7	3	3
	Wax *8	2	2
	Age resistor A *9	2	2
	Silane coupling agent *10	7	7
	Age resisto B *11	2	2
Evaluation	Vulcanization accelerator A *12	1	1
	Vulcanization accelerator B *13	1	1
	Vulcanization accelerator C *14	2	2
	Sulfur *15	2	2
Low heat generating properties		100	70
Steering stability		100	107

*1: Natural rubber, RSS #3

*2: "#80" manufactured by Asahi Carbon Co., Ltd.

*3: "Nipsil HQ-N" manufactured by Tosoh Silica Corporation

*4: "T-REZ RD104" manufactured by Tonen Chemical Corporation.

*5: Kiriin stearic acid manufactured by NOF CORPORATION

*6: "Zinc oxide 2 types" manufactured by HAKUSUI TECH CO., LTD.

*7: "Diana Process NH-70S" manufactured by Idemitsu Kosan Co., Ltd.

*8: "SUNTIGHT A" manufactured by Seiko-Chemical Co., Ltd.

*9: "Antigen 6C" manufactured by Sumitomo Chemical Co., Ltd.

*10: "ABC-856" manufactured by Shin-Etsu Chemical Co., Ltd., bis-triethoxysilylpropyl-polysulfide

*11: "NONFLEX RD-S" manufactured by Seiko-Chemical Co., Ltd.

*12: "SOXINOL D-G" manufactured by Sumitomo Chemical Co., Ltd.

*13: "SANCELER DM-TG" manufactured by SANSHIN CHEMICAL INDUSTRY CO., LTD.

*14: "NOCCELER NS-P" manufactured by Ouchi Shinko Chemical Industrial Co., Ltd.

*15: "HK200-5" manufactured by Hosoi Chemical Industry Co., Ltd.

*16: "Aktiplast ® (Aktiplast is a registered trademark in Japan, other countries, or both) PP" manufactured by LANXESS

*17: "VP1405" manufactured by HAMBURG STRUCTOL

From the results in Table 1, it is understood that the sample of Example 1 containing a modified copolymer modified with a modifier containing a compound represented by the formula (1) and having a low vinyl unit content of the modified copolymer is highly evaluated in both low

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heat generating properties and steering stability as compared with the sample of Comparative Example 1.

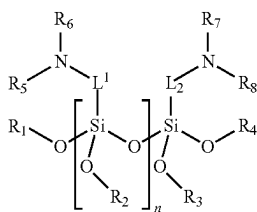
INDUSTRIAL APPLICABILITY

According to the present disclosure, it is possible to provide a rubber composition that realizes excellent low heat generating properties while maintaining good steering stability when applied to a tire. According to the present disclosure, it is also possible to provide a tire with improved rolling resistance and steering stability.

The invention claimed is:

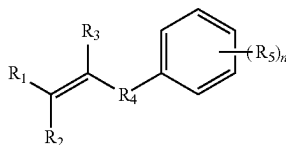
1. A rubber composition, comprising a rubber component containing a copolymer having a conjugated diene unit and an aromatic vinyl unit, and a filler, wherein

the copolymer is a modified copolymer having a content of the aromatic vinyl unit of less than 10 mass % and modified with a modifier containing a compound represented by the formula (1),



where R1 to R8 are each an independent alkyl group having 1 to 20 carbon atoms; L1 and L2 are each an independent alkylene group having 1 to 20 carbon atoms; and n is an integer of 2 to 4.

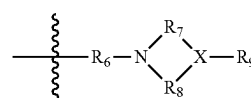
2. The rubber composition according to claim 1, wherein the copolymer is a modified copolymer further modified with a modifier containing a compound represented by the formula (2),



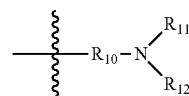
where in the formula (2), R1 to R3 are each independently hydrogen; an alkyl group having 1 to 30 carbon atoms; an alkenyl group having 2 to 30 carbon atoms; an alkynyl group having 2 to 30 carbon atoms; a heteroalkyl group having 1 to 30 carbon atoms; a heteroalkenyl group having 2 to 30 carbon atoms; a heteroalkynyl group having 2 to 30 carbon atoms; a cycloalkyl group having 5 to 30 carbon atoms; an aryl group having 6 to 30 carbon atoms; or a heterocyclic group having 3 to 30 carbon atoms, R4 is a single bond; an alkylene group having 1 to 20 carbon atoms substituted or unsubstituted with a substituent; a cycloalkylene group having 5 to 20 carbon atoms substituted or unsubstituted with a substituent; or an arylene group having 5 to 20 carbon atoms substituted or unsubstituted with a substituent, where the substituent is an alkyl group having 1 to 10 carbon atoms, a cycloalkyl group having 5 to 10 carbon atoms, or an aryl group having 6 to 20 carbon atoms, R5 is an alkyl group having 1 to 30 carbon atoms; an alkenyl group having

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2 to 30 carbon atoms; an alkynyl group having 2 to 30 carbon atoms; a heteroalkyl group having 1 to 30 carbon atoms; a heteroalkenyl group having 2 to 30 carbon atoms; a heteroalkynyl group having 2 to 30 carbon atoms; a cycloalkyl group having 5 to 30 carbon atoms; an aryl group having 6 to 30 carbon atoms; a heterocyclic group having 3 to 30 carbon atoms; or a functional group represented by the following chemical formula (2a) or chemical formula (2b), n is an integer of 1 to 5, when at least one of R5 is a functional group represented by the following chemical formula (2a) or chemical formula (2b), and n is an integer of 2 to 5, a plurality of R5s may be the same as or different from each other,



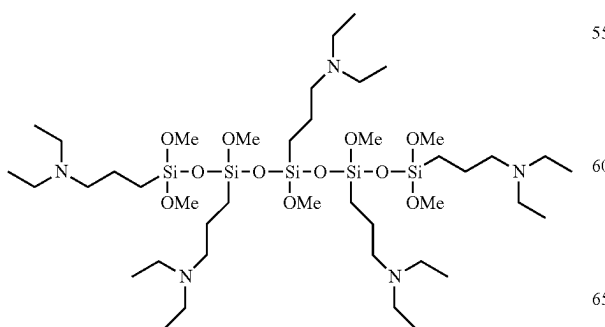
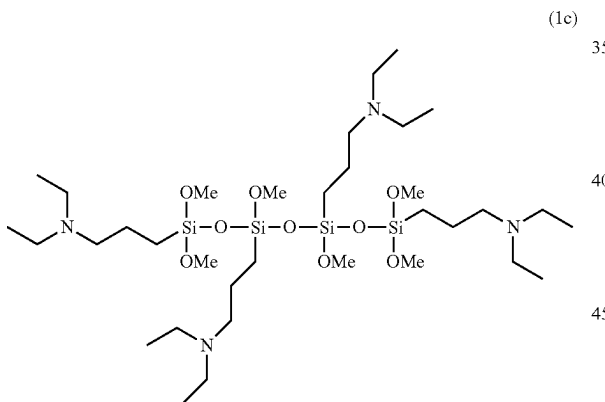
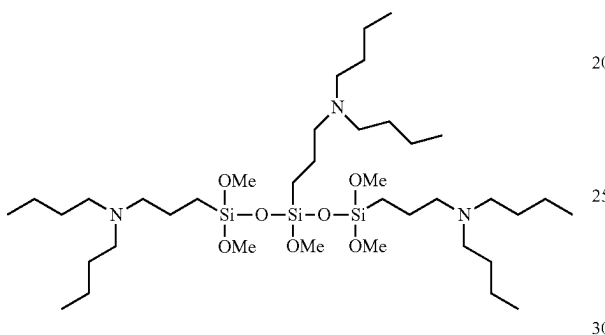
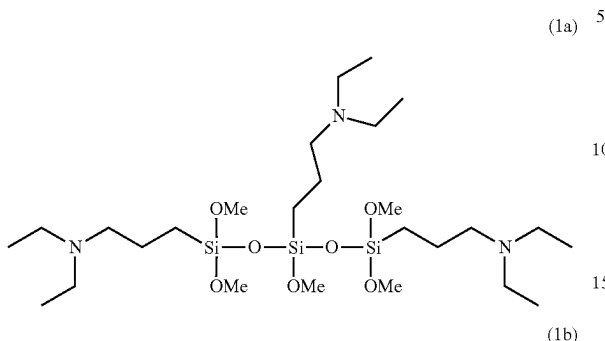
where in the formula (2a), R6 is an alkylene group having 1 to 20 carbon atoms substituted or unsubstituted with a substituent; a cycloalkylene group having 5 to 20 carbon atoms substituted or unsubstituted with a substituent; or an arylene group having 6 to 20 carbon atoms substituted or unsubstituted with a substituent, where the substituent is an alkyl group having 1 to 10 carbon atoms, a cycloalkyl group having 5 to 10 carbon atoms, or an aryl group having 6 to 20 carbon atoms, R7 and R8 are each independently an alkyl group having 1 to 10 carbon atoms, a cycloalkyl group having 5 to 10 carbon atoms, or an alkylene group having 1 to 20 carbon atoms substituted or unsubstituted with an aryl group having 6 to 20 carbon atoms, R9 is hydrogen; an alkyl group having 1 to 30 carbon atoms; an alkenyl group having 2 to 30 carbon atoms; an alkynyl group having 2 to 30 carbon atoms; a heteroalkyl group having 1 to 30 carbon atoms; a heteroalkenyl group having 2 to 30 carbon atoms; a heteroalkynyl group having 2 to 30 carbon atoms; a cycloalkyl group having 5 to 30 carbon atoms; an aryl group having 6 to 30 carbon atoms; or a heterocyclic group having 3 to 30 carbon atoms, X is a N, O or S atom, when X is O or S, R9 does not exist,



where in the formula (2b), R10 is an alkylene group having 1 to 20 carbon atoms substituted or unsubstituted with a substituent; a cycloalkylene group having 5 to 20 carbon atoms substituted or unsubstituted with a substituent; or an arylene group having 6 to 20 carbon atoms substituted or unsubstituted with a substituent, where the substituent is an alkyl group having 1 to 10 carbon atoms, a cycloalkyl group having 5 to 10 carbon atoms, or an aryl group having 6 to 20 carbon atoms, R11 and R12 are each independently an alkyl group having 1 to 30 carbon atoms; an alkenyl group having 2 to 30 carbon atoms; an alkynyl group having 2 to 30 carbon atoms; a heteroalkyl group having 1 to 30 carbon atoms; a heteroalkenyl group having 2 to 30 carbon atoms; a heteroalkynyl group having 2 to 30 carbon atoms; a cycloalkyl group having 5 to 30 carbon atoms; an aryl group having 6 to 30 carbon atoms; or a heterocyclic group having 3 to 30 carbon atoms.

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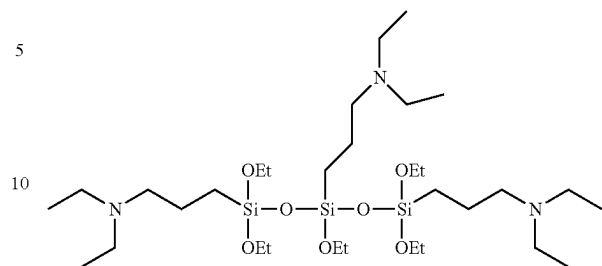
3. The rubber composition according to claim 1, wherein the modifier is any one of the formulas (1a) to (1e).



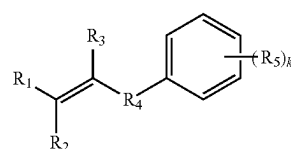
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(1e)

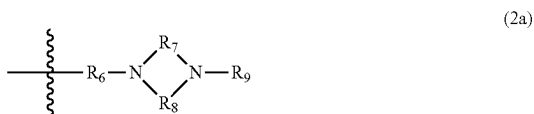


4. The rubber composition according to claim 3, wherein the copolymer is a modified copolymer further modified with a modifier containing a compound represented by the formula (2),



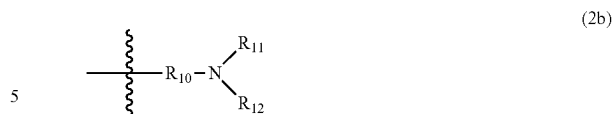
where in the formula (2), R_1 to R_3 are each independently hydrogen; an alkyl group having 1 to 30 carbon atoms; an alkenyl group having 2 to 30 carbon atoms; an alkynyl group having 2 to 30 carbon atoms; a heteroalkyl group having 1 to 30 carbon atoms; a heteroalkenyl group having 2 to 30 carbon atoms; a heteroalkynyl group having 2 to 30 carbon atoms; a cycloalkyl group having 5 to 30 carbon atoms; an aryl group having 6 to 30 carbon atoms; or a heterocyclic group having 3 to 30 carbon atoms, R_4 is a single bond; an alkylene group having 1 to 20 carbon atoms substituted or unsubstituted with a substituent; a cycloalkylene group having 5 to 20 carbon atoms substituted or unsubstituted with a substituent; or an arylene group having 5 to 20 carbon atoms substituted or unsubstituted with a substituent, where the substituent is an alkyl group having 1 to 10 carbon atoms, a cycloalkyl group having 5 to 10 carbon atoms, or an aryl group having 6 to 20 carbon atoms, R_5 is an alkyl group having 1 to 30 carbon atoms; an alkenyl group having 2 to 30 carbon atoms; an alkynyl group having 2 to 30 carbon atoms; a heteroalkyl group having 1 to 30 carbon atoms; a heteroalkenyl group having 2 to 30 carbon atoms; a heteroalkynyl group having 2 to 30 carbon atoms; a cycloalkyl group having 5 to 30 carbon atoms; an aryl group having 6 to 30 carbon atoms; a heterocyclic group having 3 to 30 carbon atoms; or a functional group represented by the following chemical formula (2a) or chemical formula (2b), n is an integer of 1 to 5, when at least one of R_5 is a functional group represented by the following chemical formula (2a) or chemical formula (2b), and n is an integer of 2 to 5, a plurality of R_5 s may be the same as or different from each other,

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where in the formula (2a), R_6 is an alkylene group having 1 to 20 carbon atoms substituted or unsubstituted with a substituent; a cycloalkylene group having 5 to 20 carbon atoms substituted or unsubstituted with a substituent; or an arylene group having 6 to 20 carbon atoms substituted or unsubstituted with a substituent, where the substituent is an alkyl group having 1 to 10 carbon atoms, a cycloalkyl group having 5 to 10 carbon atoms, or an aryl group having 6 to 20 carbon atoms, R_7 and R_8 are each independently an alkyl group having 1 to 10 carbon atoms, a cycloalkyl group having 5 to 10 carbon atoms, or an alkylene group having 1 to 20 carbon atoms substituted or unsubstituted with an aryl group having 6 to 20 carbon atoms, R_9 is hydrogen; an alkyl group having 1 to 30 carbon atoms; an alkenyl group having 2 to 30 carbon atoms; an alkynyl group having 2 to 30 carbon atoms; a heteroalkyl group having 1 to 30 carbon atoms; a heteroalkenyl group having 2 to 30 carbon atoms; a heteroalkynyl group having 2 to 30 carbon atoms; a cycloalkyl group having 5 to 30 carbon atoms; an aryl group having 6 to 30 carbon atoms; or a heterocyclic group having 3 to 30 carbon atoms, X is a N, O or S atom, when X is O or S, R_9 does not exist,

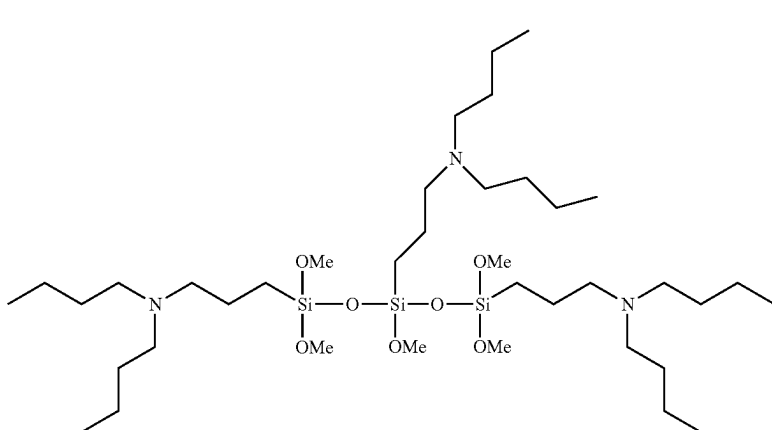
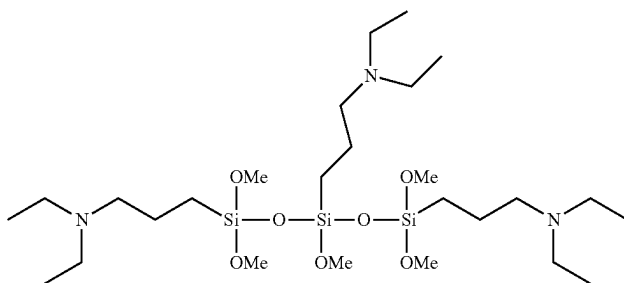
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where in the formula (2b), R_{10} is an alkylene group having 1 to 20 carbon atoms substituted or unsubstituted with a substituent; a cycloalkylene group having 5 to 20 carbon atoms substituted or unsubstituted with a substituent; or an arylene group having 6 to 20 carbon atoms substituted or unsubstituted with a substituent, where the substituent is an alkyl group having 1 to 10 carbon atoms, a cycloalkyl group having 5 to 10 carbon atoms, or an aryl group having 6 to 20 carbon atoms, R_{11} and R_{12} are each independently an alkyl group having 1 to 30 carbon atoms; an alkenyl group having 2 to 30 carbon atoms; an alkynyl group having 2 to 30 carbon atoms; a heteroalkyl group having 1 to 30 carbon atoms; a heteroalkenyl group having 2 to 30 carbon atoms; a heteroalkynyl group having 2 to 30 carbon atoms; a cycloalkyl group having 5 to 30 carbon atoms; an aryl group having 6 to 30 carbon atoms; or a heterocyclic group having 3 to 30 carbon atoms.

5. The rubber composition according to claim 1, wherein a content of the aromatic vinyl unit in the copolymer is 8 mass % or less.

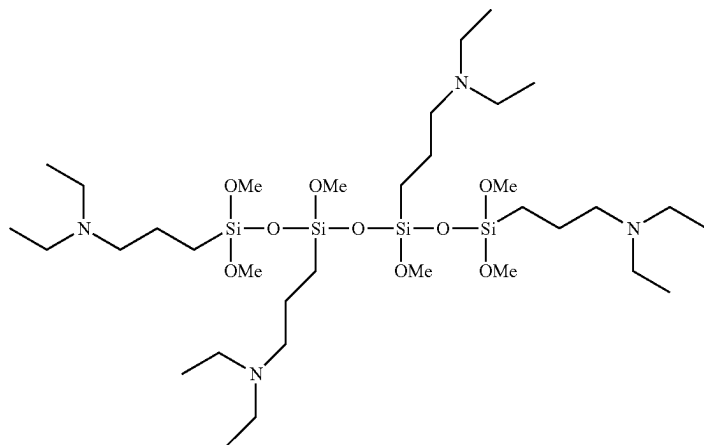
6. The rubber composition according to claim 5, wherein the modifier is any one of the formulas (1a) to (1e).



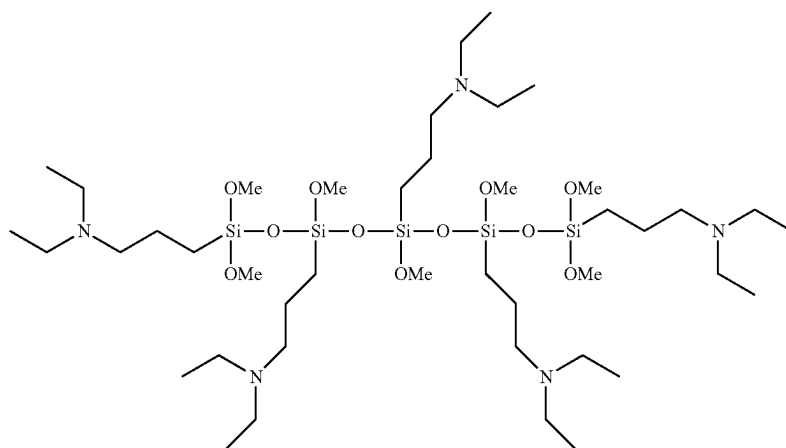
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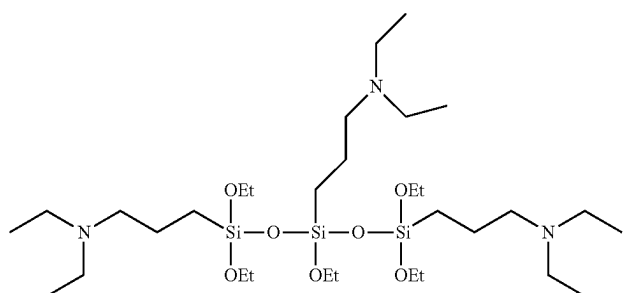
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(1c)



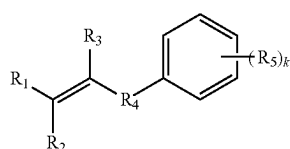
(1d)



(1e)

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7. The rubber composition according to claim 5, wherein the copolymer is a modified copolymer further modified with a modifier containing a compound represented by the formula (2),



where in the formula (2), R_1 to R_3 are each independently hydrogen; an alkyl group having 1 to 30 carbon atoms; an alkenyl group having 2 to 30 carbon atoms; an alkynyl group

having 2 to 30 carbon atoms; a heteroalkyl group having 1 to 30 carbon atoms; a heteroalkenyl group having 2 to 30 carbon atoms; a heteroalkynyl group having 2 to 30 carbon atoms; a cycloalkyl group having 5 to 30 carbon atoms; an aryl group having 6 to 30 carbon atoms; or a heterocyclic group having 3 to 30 carbon atoms, R_4 is a single bond; an alkylene group having 1 to 20 carbon atoms substituted or unsubstituted with a substituent; a cycloalkylene group having 5 to 20 carbon atoms substituted or unsubstituted with a substituent; or an arylene group having 5 to 20 carbon atoms substituted or unsubstituted with a substituent, where the substituent is an alkyl group having 1 to 10 carbon atoms, a cycloalkyl group having 5 to 10 carbon atoms, or an aryl group having 6 to 20 carbon atoms, R_5 is an alkyl group having 1 to 30 carbon atoms; an alkenyl group having 2 to 30 carbon atoms; an alkynyl group having 2 to 30 carbon atoms; a heteroalkyl group having 1 to 30 carbon

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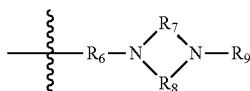
(2)

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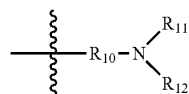
atoms; a heteroalkenyl group having 2 to 30 carbon atoms; a heteroalkynyl group having 2 to 30 carbon atoms; a cycloalkyl group having 5 to 30 carbon atoms; an aryl group having 6 to 30 carbon atoms; a heterocyclic group having 3 to 30 carbon atoms; or a functional group represented by the following chemical formula (2a) or chemical formula (2b), n is an integer of 1 to 5, when at least one of R₅ is a functional group represented by the following chemical formula (2a) or chemical formula (2b), and n is an integer of 2 to 5, a plurality of R₅s may be the same as or different from each other,



where in the formula (2a), R₆ is an alkylene group having 1 to 20 carbon atoms substituted or unsubstituted with a substituent; a cycloalkylene group having 5 to 20 carbon atoms substituted or unsubstituted with a substituent; or an arylene group having 6 to 20 carbon atoms substituted or unsubstituted with a substituent, where the substituent is an alkyl group having 1 to 10 carbon atoms, a cycloalkyl group having 5 to 10 carbon atoms, or an aryl group having 6 to 20 carbon atoms, R₇ and R₈ are each independently an alkyl group having 1 to 10 carbon atoms, a cycloalkyl group having 5 to 10 carbon atoms, or an alkylene group having 1 to 20 carbon atoms substituted or unsubstituted with an aryl group having 6 to 20 carbon atoms, R₉ is hydrogen; an alkyl group having 1 to 30 carbon atoms; an alkenyl group having 2 to 30 carbon atoms; an alkynyl group having 2 to 30 carbon atoms; a heteroalkyl group having 1 to 30 carbon atoms; a heteroalkenyl group having 2 to 30 carbon atoms; a heteroalkynyl group having 2 to 30 carbon atoms; a

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cycloalkyl group having 5 to 30 carbon atoms; an aryl group having 6 to 30 carbon atoms; or a heterocyclic group having 3 to 30 carbon atoms, X is a N, O or S atom, when X is O or S, R₉ does not exist,



(2b)

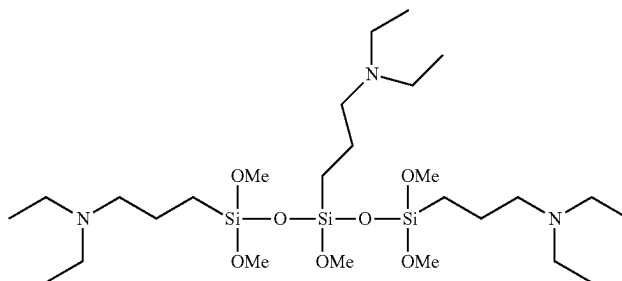
where in the formula (2b), R₁₀ is an alkylene group having 1 to 20 carbon atoms substituted or unsubstituted with a substituent; a cycloalkylene group having 5 to 20 carbon atoms substituted or unsubstituted with a substituent; or an arylene group having 6 to 20 carbon atoms substituted or unsubstituted with a substituent, where the substituent is an alkyl group having 1 to 10 carbon atoms, a cycloalkyl group having 5 to 10 carbon atoms, or an aryl group having 6 to 20 carbon atoms, R₁₁ and R₁₂ are each independently an alkyl group having 1 to 30 carbon atoms; an alkenyl group having 2 to 30 carbon atoms; an alkynyl group having 2 to 30 carbon atoms; a heteroalkyl group having 1 to 30 carbon atoms; a heteroalkenyl group having 2 to 30 carbon atoms; a heteroalkynyl group having 2 to 30 carbon atoms; a cycloalkyl group having 5 to 30 carbon atoms; an aryl group having 6 to 30 carbon atoms; or a heterocyclic group having 3 to 30 carbon atoms.

8. The rubber composition according to claim 1, wherein the rubber component further contains a conjugated diene-based rubber different from the modified copolymer.

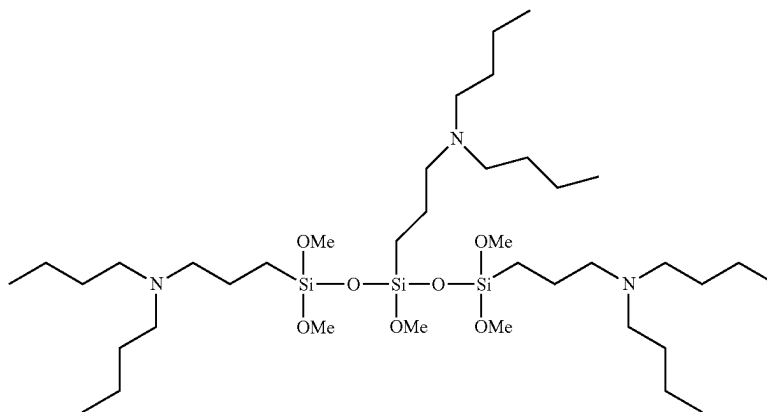
9. The rubber composition according to claim 8, wherein a content of the aromatic vinyl unit in the copolymer is 8 mass % or less.

10. The rubber composition according to claim 8, wherein the modifier is any one of the formulas (1a) to (1e).

(1a)



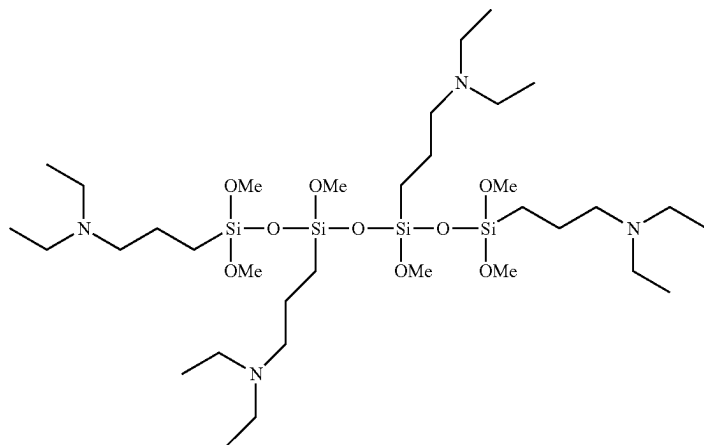
(1b)



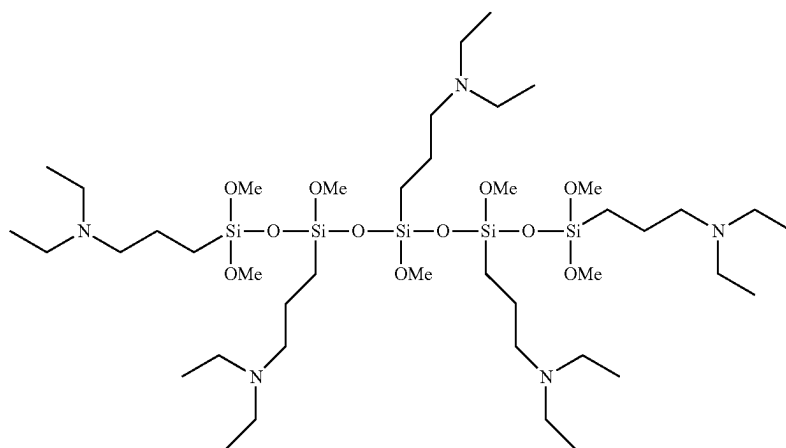
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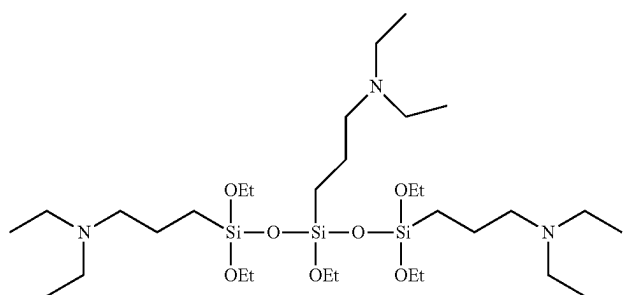
-continued



(1c)



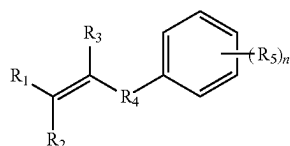
(1d)



(1e)

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11. The rubber composition according to claim 8, wherein the copolymer is a modified copolymer further modified with a modifier containing a compound represented by the formula (2),



where in the formula (2), R_1 to R_3 are each independently hydrogen; an alkyl group having 1 to 30 carbon atoms; an alkenyl group having 2 to 30 carbon atoms; an alkynyl group

having 2 to 30 carbon atoms; a heteroalkyl group having 1 to 30 carbon atoms; a heteroalkenyl group having 2 to 30 carbon atoms; a heteroalkynyl group having 2 to 30 carbon atoms; a cycloalkyl group having 5 to 30 carbon atoms; an aryl group having 6 to 30 carbon atoms; or a heterocyclic group having 3 to 30 carbon atoms, R_4 is a single bond; an alkylene group having 1 to 20 carbon atoms substituted or unsubstituted with a substituent; a cycloalkylene group having 5 to 20 carbon atoms substituted or unsubstituted with a substituent; or an arylene group having 5 to 20 carbon atoms substituted or unsubstituted with a substituent, where the substituent is an alkyl group having 1 to 10 carbon atoms, a cycloalkyl group having 5 to 10 carbon atoms, or an aryl group having 6 to 20 carbon atoms, R_5 is an alkyl group having 1 to 30 carbon atoms; an alkenyl group having 2 to 30 carbon atoms; an alkynyl group having 2 to 30 carbon atoms; a heteroalkyl group having 1 to 30 carbon

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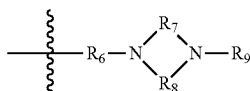
(2)

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atoms; a heteroalkenyl group having 2 to 30 carbon atoms; a heteroalkynyl group having 2 to 30 carbon atoms; a cycloalkyl group having 5 to 30 carbon atoms; an aryl group having 6 to 30 carbon atoms; a heterocyclic group having 3 to 30 carbon atoms; or a functional group represented by the following chemical formula (2a) or chemical formula (2b), n is an integer of 1 to 5, when at least one of R₅ is a functional group represented by the following chemical formula (2a) or chemical formula (2b), and n is an integer of 2 to 5, a plurality of R₅s may be the same as or different from each other,

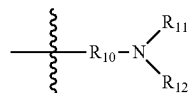


(2a)

where in the formula (2a), R₆ is an alkylene group having 1 to 20 carbon atoms substituted or unsubstituted with a substituent; a cycloalkylene group having 5 to 20 carbon atoms substituted or unsubstituted with a substituent; or an arylene group having 6 to 20 carbon atoms substituted or unsubstituted with a substituent, where the substituent is an alkyl group having 1 to 10 carbon atoms, a cycloalkyl group having 5 to 10 carbon atoms, or an aryl group having 6 to 20 carbon atoms, R₇ and R₈ are each independently an alkyl group having 1 to 10 carbon atoms, a cycloalkyl group having 5 to 10 carbon atoms, or an alkylene group having 1 to 20 carbon atoms substituted or unsubstituted with an aryl group having 6 to 20 carbon atoms, R₉ is hydrogen; an alkyl group having 1 to 30 carbon atoms; an alkenyl group having 2 to 30 carbon atoms; an alkynyl group having 2 to 30 carbon atoms; a heteroalkyl group having 1 to 30 carbon

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atoms; a heteroalkenyl group having 2 to 30 carbon atoms; a heteroalkynyl group having 2 to 30 carbon atoms; a cycloalkyl group having 5 to 30 carbon atoms; an aryl group having 6 to 30 carbon atoms; or a heterocyclic group having 3 to 30 carbon atoms, X is a N, O or S atom, when X is O or S, R₉ does not exist,



(2b)

where in the formula (2b), R₁₀ is an alkylene group having 1 to 20 carbon atoms substituted or unsubstituted with a substituent; a cycloalkylene group having 5 to 20 carbon atoms substituted or unsubstituted with a substituent; or an arylene group having 6 to 20 carbon atoms substituted or unsubstituted with a substituent, where the substituent is an alkyl group having 1 to 10 carbon atoms, a cycloalkyl group having 5 to 10 carbon atoms, or an aryl group having 6 to 20 carbon atoms, R₁₁ and R₁₂ are each independently an alkyl group having 1 to 30 carbon atoms; an alkenyl group having 2 to 30 carbon atoms; an alkynyl group having 2 to 30 carbon atoms; a heteroalkyl group having 1 to 30 carbon atoms; a heteroalkenyl group having 2 to 30 carbon atoms; a heteroalkynyl group having 2 to 30 carbon atoms; a cycloalkyl group having 5 to 30 carbon atoms; an aryl group having 6 to 30 carbon atoms; or a heterocyclic group having 3 to 30 carbon atoms.

12. A tire, using the rubber composition according to claim 1.

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